

Towards Capturing the Therapeutic Effects of Music in Everyday Life: Building the Beats Dataset

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Abstract

Personalised music therapy shows promise as a safe, cost-effective and non-intrusive alternative to pharmacological interventions to support health and wellbeing. However, artificial lab settings limit replication in real-life scenarios and obscure which musical features consistently elicit specific physiological responses (e.g., heart rate) in real environments, we present a framework that leverages commercially available music streaming and wearable sensing technologies to enable continuous, in-the-wild physiological data collection and real-time music tracking. As part of the study design, therapeutically targeted music was systematically selected and listened to in real-world contexts over a sustained experimental period. The proposed system monitors and aligns music listening activity with physiological data and self-reported effects, as well as capturing listeners' demographic information, individual traits, and pre-/post-experiment wellbeing measures. We detailed our stimuli selection process, followed by an overview of the system components. Crucially, this research lays the groundwork for generating a first-of-its-kind dataset to study how music truly influences our physiology in everyday life, offering insights that could guide the development of more effective personalised music-based interventions.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

Wearable Sensing, In-the-Wild Data Collection, Physiological Signals, Affective Computing, Music Therapy

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1 Introduction

The role of music in supporting health and wellbeing is steadily gaining strong research attention as music is increasingly integrated into people's lives. Recent data show that around 75% of people in a global survey listen to music to reduce stress, while the volume of research publications on music and health has tripled over the last decade [7, 8]. Evidence from clinical studies reports the benefits of music interventions in managing a broad spectrum of conditions, ranging from neurodegenerative disorders and psychiatric conditions to pain and stress-related symptoms [1, 3, 6, 18]. These healing effects have been attributed to music's ability to influence neurophysiological activity, which music therapists harness to regulate arousal and support physical functions. [4]. Despite growing empirical support, the mechanisms underlying music's therapeutic effects on the brain and body remain only partially understood. Amid these unresolved questions, researchers have turned to physiological signals as objective, real-time measures of emotional responses, surpassing the reliability of self-reports and behavioural observations.

Measures such as Photoplethysmography (PPG) and Electrodermal Activity (EDA) are commonly used to infer valence and arousal, respectively, providing objective measures of emotional change over time. While the automatic recognition of valence and arousal levels via physiological indicators has been highlighted as a novel direction [26], expanding this to include their regulation through targeted interventions could further enhance healthcare applications. For music to function as an effective regulatory tool, establishing consistent links between musical elements and physiological responses is essential to achieving predictable therapeutic effects. Therefore, a reliable and reproducible taxonomy of therapeutic audio features with corresponding physiological correlates could guide the design of more effective music-based interventions.

However, replicating results across studies remains a central challenge, preventing a unified understanding of which musical features consistently elicit specific physiological responses [13]. This is further complicated by the dominance of lab-based research, which likely suppresses natural responses to music [15], and by the lack of publicly available datasets that offer music-aligned physiological data collected outside controlled environments. Consequently, the use of music to support physical and emotional health outside clinical settings remains a complex task, requiring sustained validation of its consistent effects across diverse real-world contexts and populations. Fortunately, recent advancements in wearable technology and music streaming platforms offer new opportunities to capture realistic physiological responses to acoustic stimuli, enabling in-the-wild data collection at an unprecedented scale [10].

In response to the outlined challenges and the need for ecologically valid approaches, we present an integrated solution currently used to collect a real-world music dataset with aligned physiological signals. The system combines music streaming and wearable technologies, along with a custom-made API and a study portal to facilitate participant recruitment and survey administration. Participants are recruited in batches and take part in a three-week study in which they use the wearable device continuously, listen to curated therapeutic playlists, and self-report whether these produced the intended effect. Accordingly, the value of the presented work is twofold: (1) we present a systematic approach for the selection of therapeutic music offered by streaming services, and (2) we describe the main components of the system to capture music’s physiological effects in everyday settings.

2 Related Work

Table 1 presents a summary of datasets offering aligned physiological signals during music listening, outlining key characteristics that served as reference points to guide the development of the present system. We briefly describe available music-evoked datasets, focusing initially on their main features, followed by a discussion of common limitations. For a more exhaustive review of available music datasets, please refer to [19].

Perhaps the most influential datasets is DEAP [20], one of the most widely used benchmarks, offering multimodal physiological signals, front-facing video data and self-reports collected as participants were exposed to one-minute music video clips. In contrast to DEAP, the PMemo dataset [28] includes EDA responses from multiple participants to chorus excerpts of popular music, along with dynamic and static self-reports and song metadata.

In comparison, the HKU956 dataset [14] offers multimodal physiological signals and self-reports in response to copyright-free music, while also capturing individual differences such as personality and demographics. Distinct from the datasets already mentioned, the SiTunes dataset [11] stands out for its real-world data collection, incorporating contextual variables and listener states before and after music exposure. Nevertheless, physiological data is limited to heart rate, and the data collection instrument is unspecified.

Notably, the datasets range from multimodal, encompassing subjective ratings, along with behavioural and physiological data, to unimodal sets focused on one signal. Such unimodal datasets may overlook the dual dimensions of emotions, which involve both valence and arousal components. Additionally, using popular song excerpts with vocals and music videos makes it difficult to isolate effects driven solely by musical features.

A further limitation is the general absence of measures that capture individual differences known to shape emotional responses to music, with only one dataset including information on personality traits. Most importantly, a notable gap remains in datasets captured under real-world conditions, with most studies relying on constrained laboratory settings that may fail to capture realistic responses to music. As emphasised in music psychology research [17], the emotional impact of music is shaped by contextual factors such as musical function, familiarity, and listener level of choice. These effects are further influenced by individual differences, yet such factors are largely absent in currently available music-evoked physiological datasets.

3 Methodology

We currently use the proposed system to collect data from participants in free-moving environments over three weeks using a research-grade wearable device. This section outlines the systematic approach to selecting therapeutic music stimuli and describes the main components that support the ongoing data collection.

3.1 Stimuli Selection

Our stimuli selection process was driven primarily by empirical findings from music psychology literature focusing on identifying functions of music for enhancing wellbeing [12]. Given the wide range of music functions for therapeutic purposes, we narrowed our selection by concentrating on three higher-order music functions:

- **Goal Attainment** — music supporting motivation, task enjoyment, and persistence (e.g., for work or study).
- **Affect Regulation** — music for mood improvement, stress relief, and emotional engagement (e.g., for relaxation).
- **Eudaimonic** — music facilitating introspection, meditation, and personal meaning (e.g., for mental clarity).

We then turned to Spotify¹ as our source of music selection due to its extensive curated playlist catalogue.

We restricted our selection by contemplating only official Spotify editorial playlists, as these are crafted by editors with domain expertise and further enhanced by data-driven insights [2]. Additionally, we excluded music with lyrics to minimise language and cultural bias. We prioritised playlists with a significant number of followers featuring simple arrangements with few instruments to facilitate acoustic analysis. Based on these criteria, we systematically selected therapeutic music by matching keywords from music functions to support well-being with the playlist’s description while also prioritising popularity and relevant attributes previously stated.

To select each of the playlists i in the study from Spotify’s editorial collection, each was assigned a music function F_{music} (Eq. 1) corresponding to its highest weighted relevance score WRS (Eq. 2) across all higher-order music functions in J [12].

$$F_{music}(i) = \arg \max_j (WRS_{i,1}, WRS_{i,2}, \dots, WRS_{i,j}) \quad (1)$$

$$WRS_{i,j} = w_1 P_i + w_2 S_{c_i} + w_3 S_{f_i} \quad (2)$$

Where P_i is an indicator of popularity represented by the log normalisation of the number of followers of each playlist, S_{c_i} corresponds to the average of the cosine similarity scores between the description of the playlists and custom keywords defined for simplicity and lack of vocals, and S_{f_i} denotes the average of the cosine similarity scores between the description of the playlists and keywords associated with each function of music j .

Both S_{c_i} and S_{f_i} account for the semantic similarity between keywords and the description of playlists, where the pre-trained model *all-mpnet-base-v2*² was utilised for embedding extraction. The values assigned to w_1 , w_2 and w_3 are 0.3, 0.3 and 0.4, respectively. Then, one designates the function of music for each playlist by selecting the function with the highest WRS , and the most suitable playlist for each function is selected based on the highest score.

¹<https://www.spotify.com/us/>

²<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

Table 1: Summary of available music datasets with aligned physiological data.

Dataset	Stimuli	Participants	Data Collected	Devices	Settings
DEAP[20]	40 one-minute music videos	32	EEG, GSR, EMG, BVP, face video (22), self-reports (valence, arousal, liking, dominance, familiarity)	Biosemi ActiveTwo	Controlled lab
PMemo[28]	794 pop chorus excerpts (varied duration)	457	EDA, static/dynamic valence-arousal ratings	Biopac MP150	Controlled lab
HKU956[14]	956 full-length CC-licensed songs, ~50 sec avg.	30	BVP, EDA, skin temperature, valence/arousal ratings	Empatica E4	Controlled lab
SiTunes[11]	307 music tracks (varied duration)	30	HR, activity, location, valence/arousal, liking, duration	Smart wristband (unspecified)	One-week field study

As a final step, we refined our selection to make the task more manageable by subsampling the most popular songs from the original playlists. This resulted in twelve playlists, four for each of the three music functions. Each playlist lasts approximately 30 minutes, with 119 songs in total across the twelve playlists.

3.2 System Overview

Data collection in this study relies on multiple essential materials to capture psychometrics, physiological signals, listening activity, and self-reports. Participants wear a research-grade wearable device continuously for 21 days, engaged with curated playlists during their daily routines, and provided feedback via short surveys. Figure 1 offers a schematic overview of the system architecture, core components and data collected³.



Figure 1: Overview of platform key elements enabling data collection from music listeners in free-moving environments over an extended period of time.

3.2.1 Questionnaires. Findings from previous research have suggested that emotional response to music could be moderated by personality [16], music training [22] and music preference [24].

³Some icons sourced from Flaticon.

These individual factors are obtained by means of the Ten Item Personality Inventory (TIPI) [9], Gold-MSI [23] and the STOMP [25], respectively. Likewise, including therapeutic music in participants' daily lives motivated the assessment of well-being through pre- and post-experiment measures. Therefore, participants report on their mood, perceived stress and depression severity with the PANAS [27], PSS [5] and PHQ-9 [21] questionnaires at both the start and end of the experiment.

3.2.2 Wearable Device and Companion App. The EmbracePlus⁴ wristband and its companion Care App⁵ are used for continuous physiological monitoring in ambulatory settings to collect blood volume pressure (BVP), electrodermal activity (EDA), skin temperature and triaxial acceleration. Due to its unobtrusive nature and medical-grade sensors, the device enables seamless, long-term physiological monitoring in ambulatory settings without disrupting participants' daily activities. Data is continuously transmitted via Bluetooth to the paired mobile phone, which securely uploads it to Empatica Cloud. To maximise data collection, participants are encouraged to wear the wristband at all times while ensuring the application remains active in the background.

3.2.3 Spotify Accounts. We created dedicated Spotify accounts for this study. Playlists' names, descriptions, and cover images were customised to facilitate the identification of their intended therapeutic benefits. All twelve playlists were added to each account's library and organised into directories based on their music function. The accounts' settings are adjusted for each participant to guarantee recording reliable playback data. Moreover, autoplay similar content is also disabled to prevent playing foreign songs when participants finished listening to the study's playlists. Similarly, Smart Shuffle is discouraged to avoid adding unrelated tracks to the queue while participants engages with the therapeutic playlists.

3.2.4 Real-time Music Tracking and Physiology Alignment. Real-time playback tracking was automated through periodic jobs using the Spotify API. Playback data is retrieved every 30 seconds, with

⁴<https://www.empatica.com/en-gb/embraceplus/>

⁵<https://support.empatica.com/hc/en-us/articles/9843547409821-Download-the-Care-App>

logged timestamps cross-validated against Spotify records to ensure accuracy. This application is hosted on university servers and aligns physiological data with music-listening activity while securely saving them in cloud storage. Physiological data is pulled daily directly from Empatica Cloud, and only tracks listened to within the corresponding timeframe are included. The software works as a centralised system, linking all accounts to facilitate music listening tracking without participants needing to install additional apps. However, for accurate tracking, a stable internet connection is required.

3.2.5 Self-Report Survey. An automated survey distribution mechanism was also integrated into the system. This module sends an email notification to participants who listened to the study's playlists within the last twelve hours, prompting them to respond to a brief confirmatory question assessing the perceived helpfulness of the playlists. Subjective responses are linked to physiological and playback data using a session ID, which associates participants' feedback with the specific listening session of the therapeutic playlist. This approach enables complementing objective data with subjective reports, allowing confirmation or contextualisation of patterns observed in physiological measures when listening to music.

3.2.6 Data Records. All participant data is anonymised and stored in dedicated directories, with records including demographic (e.g., gender, age), individual differences (e.g., personality traits, music preferences), and pre/post-experiment wellbeing measures (e.g., stress, depression). It also includes music listening activity, raw physiological data (BVP and EDA indexing affective states), and self-reports evaluating if the playlists produced the intended effect.

4 Discussion & Future Work

The system has been operating reliably, prompting us to share this early-stage work with the research community. However, to move this research forward, certain limitations should be addressed.

Spotify's therapeutic playlists supports a narrow set of scenarios with limited musical diversity, which may reduce participant engagement. Additionally, the use of study-specific Spotify accounts restricts access to users' personalised playlists, reducing convenience. At the system level, reliance on third-party tools such as Spotify and Empatica introduces scalability challenges. Larger participant cohorts risk exceeding API quotas, and Empatica's academic licensing limits each account to a single study, constraining cross-institutional collaboration. Addressing these constraints, through service upgrades, could help solve scalability limitations.

Future work will focus on curating and releasing the dataset, followed by analysis to identify consistent physiological patterns across individuals and map them to therapeutic acoustic features for intervention design. Particular focus will be placed on generating adaptive auditory patterns to moderate arousal levels. Additionally, we plan to conduct semi-structured interviews with a subset of participants to explore individual experiences and contextualise observed physiological responses. Lastly, the development of a dedicated mobile app is being explored to streamline API usage and enable richer tracking of music listening sessions.

5 Conclusion

This ongoing work contributes to the growing effort to understand how music influences physiology in real-world contexts. We presented an integrated system building on gaps identified in existing datasets and lab-based studies. The system combines therapeutic music selection via streaming platforms, wearable sensing, and participant self-reporting in a three-week field deployment. While data collection is ongoing, early use has demonstrated system stability and participant active engagement. We discussed practical limitations, including third-party dependencies and limited availability of therapeutic playlists, which may affect scalability. As future steps, we plan to curate and publicly release the dataset under collection, conduct both quantitative and qualitative analyses, and explore the development of a mobile application to optimise API usage and support richer data acquisition of music sessions.

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